
Categories of Solids

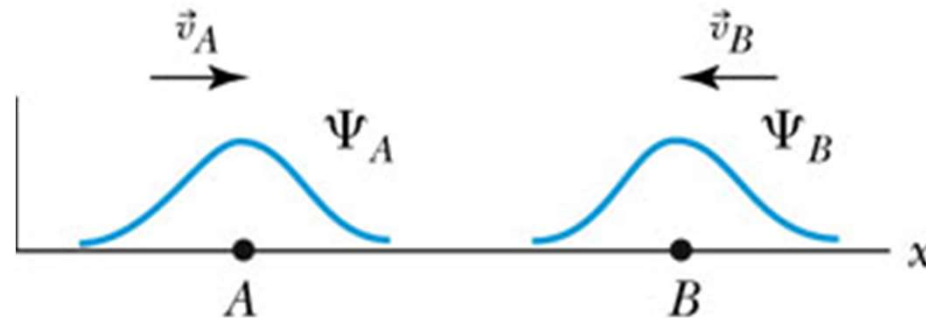
- There are three categories of solids, based on their **conducting properties**:
 - conductors
 - semiconductors
 - insulators

Band Theory of Solids

- In order to account for *decreasing* resistivity with increasing temperature as well as other properties of semiconductors, a new theory known as the **band theory** is introduced.
- The essential feature of the band theory is that the allowed energy states for electrons are nearly continuous over certain ranges, called **energy bands**, with forbidden energy gaps between the bands.

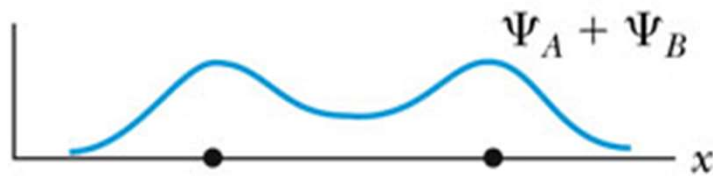
Band Theory of Solids

- Consider initially the known wave functions of two hydrogen atoms far enough apart so that they do not interact.



Band Theory of Solids

- Interaction of the wave functions occurs as the atoms get closer:



Symmetric



Antisymmetric

- An electron in the symmetric state has a nonzero probability of being halfway between the two atoms, while an electron in the antisymmetric state has a zero probability of being at that location.

Band Theory of Solids

- In the symmetric case the binding energy is slightly stronger resulting in a lower energy state.
 - Thus there is a splitting of all possible energy levels (1s, 2s, and so on).

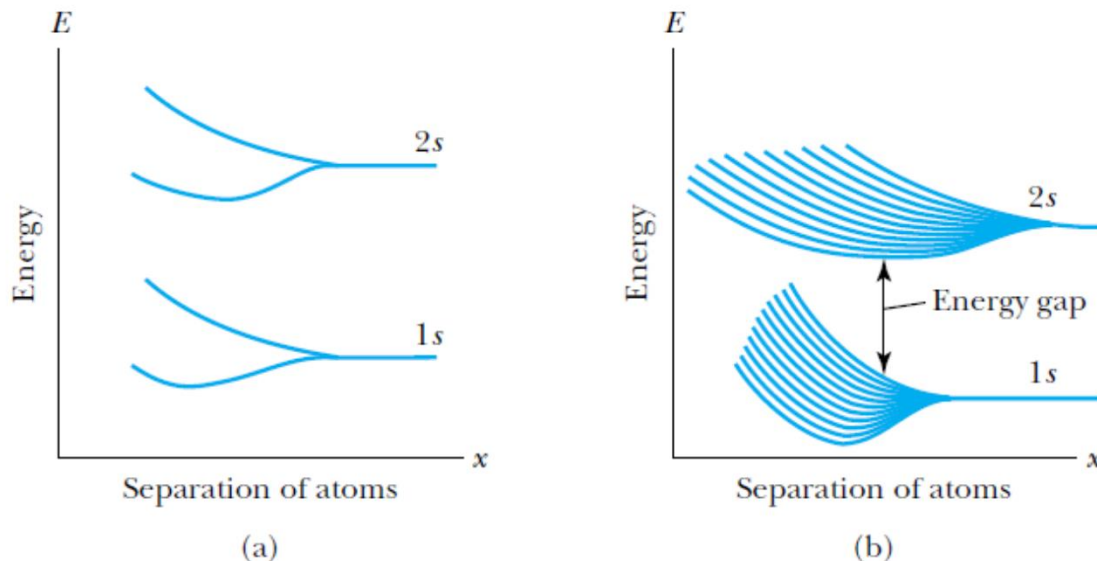


Figure 11.3 The 1s and 2s energy-level splittings of approaching hydrogen atoms for (a) 2 atoms and (b) 11 atoms. Notice the splitting of each energy level into a nearly continuous band.

When more atoms are added (as in a real solid), there is a further splitting of energy levels. With a large number of atoms, the levels are split into nearly continuous energy bands, with each band consisting of a number of closely spaced energy levels.

